GAI Empowers The Inheritance, Innovation, And Risk Response Of Intangible Cultural Heritage

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Abstract. Generative artificial intelligence (GAI) has opened up a new path for the digital inheritance and innovative expression of intangible cultural heritage, significantly reducing the threshold for public participation and stimulating the potential for human-machine collaborative creation. However, technological intervention also carries multiple structural risks such as cultural distortion, algorithmic bias, embodied dissolution, and ambiguous ownership. Single technological optimization is difficult to achieve, and there is an urgent need to establish a systematic governance framework. We should strengthen the technological bottom line through the construction of an exclusive knowledge base and a trustworthy generation mechanism for intangible cultural heritage, establish ethical norms, explore collective property rights, and promote cultural impact assessment to improve institutional guarantees. We should empower inheritors, establish their confirmation rights, and promote community governance to reshape their central position as the main body. Only by adhering to the principle of "people-oriented and cultural authenticity" can we achieve the dynamic continuation and creative transformation of intangible cultural heritage in the era of intelligence.

ACM CCS (2012) Classification: Computing methodologies → Artificial intelligence → Natural language processing → Discourse, dialogue and pragmatics

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1. Introduction

Intangible cultural heritage carries the historical memory and cultural genes of a nation, and its active inheritance has long been limited by multiple challenges such as a single mode of dissemination, alienation among young people, and outdated recording methods. The wave of technology is surging, and the rapid development of generative artificial intelligence (GAI) technology represented by big language models has opened up new paths for the digital protection and innovative expression of intangible cultural heritage. This technology, with its multi-modal generation capability, can transform traditional skills, oral narratives, and festival ceremonies into images, videos, or interactive content, significantly reducing the threshold for public participation and promoting open and collaborative cultural production.

However, there are hidden risks behind technological empowerment. When algorithms become a new form of cultural interpretation, who will safeguard the authenticity of tradition? Generative models rely on massive network data for training, and such data often lacks accurate representation of the cultural context, local knowledge, and spiritual core of intangible cultural heritage. The resulting algorithm bias may cause AI output to deviate from cultural authenticity, and even lead to symbol misreading, historical misplacement, or style homogenization. What is even more alarming is that technological systems often use the pretext of "neutrality" and "objectivity" to weaken human value judgments, but in reality conceal mainstream cultural preferences, marginalize minority or niche intangible cultural heritage projects, and form a new type of digital cultural inequality[1]. Although current research has focused on the potential application of AI in the field of intangible cultural heritage, it mostly focuses on technological implementation and dissemination effectiveness, and the systematic analysis of risk mechanisms is still insufficient. Some practices regard AI as a pure tool and overlook its role as a "cognitive mediator" in reconstructing the process of cultural meaning generation[2]. When the generated content is detached from the main body of the inheritor and driven solely by algorithmic logic, the core dimensions of authenticity, embodiment, and cultural identity of intangible cultural heritage are easily dissolved.

2. RelatedWork

Understanding the basic logic of generative artificial intelligence empowering intangible cultural heritage is a prerequisite for recognizing its risks. The intervention of generative artificial intelligence in intangible cultural heritage is achieved by reconstructing the underlying logic of cultural production and cognition, pushing intangible cultural heritage from a closed experience inheritance to an open intelligent collaborative ecosystem. Its core lies in dynamic generation and diverse co creation. Firstly, large models possess strong cross modal semantic understanding and translation capabilities. The complex cultural symbols such as patterns of traditional handicrafts, rhythms of oral epics, and spatial layouts of festive rituals can be transformed into textual prompts, which can then generate images, 3D models, or interactive scenes. This process lowers the threshold for non professional groups to understand and participate in intangible cultural heritage, making cultural expression more accessible[2]. The "Folk-rnn" project by Irish scholars utilizes artificial intelligence tools to learn 23000 traditional dance songs and establish a large-scale model. It dynamically generates new melodies that conform to cultural grammar and is open for global musicians to play and adapt, forming a dynamic inheritance model. This significantly reduces the threshold for public participation in intangible cultural heritage and reflects the empowering value of GAI in cultural production. Secondly, generative systems support the creative production mode of human-machine collaboration.

Besides, generative systems support the creative production mode of human-machine collaboration. The inheritance of intangible cultural heritage is no longer limited to face-to-face teaching between master and apprentice, but can be achieved through intelligent tools for division of labor and collaboration. Designers can input prompts such as "traditional patterns+modern style", and AI can quickly generate dozens of derivative schemes that conform to the process logic. Inheritors can select and manually implement them from them. These techniques are particularly applied in intangible cultural heritage such as sculpture, rubbings, and patterns that can innovate traditional patterns. During this process, AI undertakes repetitive drawing and color matching tasks, while humans focus on aesthetic judgment and cultural adaptation, forming efficient division of labor. Furthermore, the large model can also simulate counterfactual historical scenarios, providing inspiration and reference for innovation[3].

Finally, intelligent tools significantly lower the threshold for public participation. Ordinary users can generate emoticons, posters, or short videos with intangible cultural heritage elements by calling AI interfaces through social media platforms, and share them with just one click. This type of content, due to its convenient production and innovative form, is easy to spontaneously spread among young people, effectively alleviating the dilemma of the disappearance of intangible cultural heritage.

Indeed, all technological applications must be confirmed by humans as the ultimate boundary. “We must resist the deprivation of humanities research in computer science and related fields”, As Klein pointed out, who is Winship Distinguished Research Professor and Associate Professor in the departments of Data & Decision Sciences and English at Emory University, "Models make words, but people make meaning" [4]. The value of AI lies not in replacing the inheritor, but in liberating its creativity, making it more focused on emotional transmission, ethical judgment, and the protection of the spiritual core.

3. The Core Risks Of GAI Intervention In Intangible Cultural Heritage

Generative artificial intelligence, while empowering the inheritance of intangible cultural heritage, also introduces a series of complex and intertwined risks. These risks not only manifest as technical inaccuracies, but also delve deeper into structural issues such as cultural authenticity, social equity, subject identity, and institutional deficiencies. If there is a lack of systematic identification and effective regulation, technological dividends may transform into cultural crises. The basic logic of empowering intangible cultural heritage based on generative artificial intelligence categorizes core risks into the following five dimensions.

3.1. Digital Drive Leads To Distortion Of Cultural And Historical Context

The generative model relies on large-scale network data training, while the Internet content is generally fragmented, entertaining and de contextualized. When such data becomes the main source of intangible cultural heritage knowledge, AI is prone to generating "pseudo traditional" information that is detached from its native cultural soil. For example, in the generative creation of intangible cultural heritage themes, models may exhibit temporal cultural symbol mismatch adaptation, or simplify sacrificial scenes with specific ritual attributes into purely decorative visual elements, thereby dissolving the ritual logic and symbolic system behind them. This distortion is not an accidental error, but rather stems from the algorithm's preference for the high-frequency processing mode of generative artificial intelligence—— the more widely disseminated the simplified symbols are, the easier it is to reinforce the output.

Even with the introduction of Retrieval Augmented Generation (RAG) technology that allows models to "first search for external information and then generate answers" to solve problems such as large model illusion, knowledge lag, and lack of domain specificity by accessing authoritative databases, risks have not been eradicated. If there are errors or incompleteness in the retrieval source itself, the model will still reason based on incorrect premises. More seriously, large models have a highly fluent "illusion" ability, capable of generating seemingly reasonable but actually fictional content with a confident tone. If the public lacks professional judgment, it is easy to mistake AI output for true tradition. For example, a certain AI platform once generated the Tang Dynasty Suzhou embroidery work "Double-sided Embroidered Cat" in China, but Suzhou embroidery had not yet formed a mature system in the Tang Dynasty. If such time misalignment is not corrected in a timely manner, it will mislead the public's historical cognition.

3.2 Big Data: Data Colonization and Cultural Power Imbalance

The current mainstream training data for large models is highly concentrated in national and regional standard languages, while local dialects, minority languages, and related cultural practices have a very low proportion in the data pool. This leads to AI being unable to generate effective content or forcibly applying mainstream cultural templates when dealing with peripheral intangible cultural heritage projects such as Dong ethnic songs, Yi ethnic lacquerware, and Hezhe ethnic fish skin clothing. This kind of technology that appears to be universal, but actually replicates and amplifies existing cultural inequality, is actually a phenomenon of "data colonization". “Data colonization" is an important theory proposed by Nick Couldry, the Professor of Communication Studies in the UK, teaching at Goldsmiths College, University of London, in the era of modern computing technology. Its core lies in the collection and commodification of personal data, and through technological means such as digital platforms, it fully integrates daily life into the capitalist economic system[5].Given this, Gallegos believes that there is an urgent need for a comprehensive and systematic strategy to address this issue [6]. Algorithm bias is also reflected in the algorithm recommendation mechanism. The platform, based on artificial intelligence and big data, tends to push content that is visually impactful, fast-paced, and easy to spread in short videos to contemporary people, while ignoring slow paced intangible cultural heritage that requires immersive experience, such as Chinese Peking Opera and the African Mari é l é d music of the Kola Qin. As a result, inheritors are forced to conform to algorithmic logic, simplifying complex skills into instant popular science, sacrificing the depth of intangible cultural heritage to gain traffic. Over time, the diversity of intangible cultural heritage has been compressed into a few templates, and local knowledge systems are facing systematic dissolution[7].

3.3 Virtual Practice&Reduce Embodied Cognition

The core of intangible cultural heritage lies in the physical practice of 'learning by doing'. Most intangible cultural heritage relies on long-term interaction and practical knowledge accumulation between master and apprentice. AI generated tutorials often break down skills into linear steps or static images, stripping away the dynamic feedback between materials, environment, and body. Although learners can imitate the appearance, it is difficult for them to grasp the deep connotations of intangible cultural heritage. What is even more alarming is that virtual interaction is replacing real contact. Users try on ethnic costumes through AR and use AI to synthesize folk songs for singing, which enhances their sense of participation but also weakens their respect and support for the real inheritors. When cultural experiences are simplified into clicks on the screen, the public's emotional investment in intangible cultural heritage tends to be superficial. The inheritor is no longer a cultural authority, but has become a 'background material provider'. This alienation of relationships fundamentally shakes the foundation of intangible cultural heritage as a living practice.

3.4 The Issues Of Intellectual Property And Cultural Sovereignty

The copyright ownership of intangible cultural heritage derivatives created by generative artificial intelligence has not been precisely defined, making them orphan works that cannot be accurately defined[9].Are you the prompt word input person? Model developer? Platform operator? Or is it a cultural source community? The current AI generated content is often jointly completed by humans and machines, with blurred boundaries. Commercial institutions generate a large number of images and symbols for product design without paying licensing fees to the inheritor community, which constitutes a de facto issue of creative ownership. Even more serious is the misuse of culturally sensitive content. Some AI drawing tools allow users to freely generate images with Tibetan Buddhist symbols and shaman masks for game skins or advertising backgrounds. Such behavior not only violates the seriousness of religion, but also infringes on the cultural sovereignty of ethnic groups.

3.5 Misunderstanding Of Historical Authenticity

Generative models have powerful counterfactual reasoning abilities and can simulate hypothetical scenarios such as "if a certain skill has not been lost" or "if a certain festival continues to this day". Although such simulations have enlightening value, there are still systematic flaws in the logical consistency and historical authenticity of large models. For example, AI may generate "blue and white porcelain designed by AI in the Song Dynasty", but ignore the historical fact that blue and white cobalt materials were only introduced to China in the Yuan Dynasty. The problem is that once such highly realistic fictional content enters the public communication field, it is easily mistaken for a "new discovery" or "restoration result". Especially on short video platforms, AI generated content lacking annotations often spreads under the guise of traditional cultural popularization, blurring the boundary between reality and imagination. After repeated exposure, the public may internalize these fictional narratives as collective memories. If the human confirmation process is abandoned, technology will dominate cultural interpretation and deviate from the original intention of intangible cultural heritage protection.

4 GAI's Risk Response Path

Faced with the multiple risks caused by the intervention of generative artificial intelligence in intangible cultural heritage, technological repairs or moral appeals are no longer effective. It is necessary to establish a collaborative governance system that covers technology design, institutional regulation, and subject empowerment. This system takes cultural authenticity as the bottom line, inheritor subjectivity as the core, and human-machine collaborative verification as the operating principle, fundamentally realizing the risk management brought by artificial intelligence during generation.

4.1. Technology Governance

Technology itself is not value neutral, and its architectural design deeply influences the direction of cultural expression. Therefore, risk prevention and control must be prioritized in the model development and data preparation stages.

4.1.1.Building an exclusive corpus and model for intangible cultural heritage

The general big model relies on the open data of the Internet, which will inevitably mix noise and prejudice. By collecting first-hand information such as oral history of inheritors, craft archives, local chronicles, and ritual videos through the system, a high-quality and contextually relevant cultural database can be constructed. On this basis, utilizing entity recognition and relationship extraction techniques, a multidimensional correlation network of "skills materials rituals beliefs" is established. When AI generates content, it can call the knowledge graph in real time for logical verification to avoid era misalignment or symbol misuse[2]. For example, when generating sculptures from the Renaissance period, the system automatically checks whether the costumes and architectural forms of the characters comply with the regulations of that time.

4.1.2.Promote Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback is a machine learning paradigm that combines reinforcement learning with human subjective preferences, with the core goal of making the model output more in line with human values, intentions, and aesthetic needs. The traditional generation process is completed by the user inputting prompt words and the model outputting the results. The RLHF mechanism introduces inheritors as "cultural judges" to score the generated samples, and the model iteratively optimizes based on this. This process internalizes expert knowledge into algorithm preferences, enabling technology to truly serve the preservation of cultural authenticity. It is divided into three stages:

1. Supervised Fine Tuning (SFT) first uses annotated high-quality human demonstration data to fine tune the pre trained large model, allowing the model to learn human expression habits and task requirements, and generate output that meets basic expectations. For example, let the model learn professional expressions of "Introduction to Intangible Cultural Heritage Instruments" instead of colloquial and fragmented content.

2. Reward Model Training (RMT) allows the model to generate multiple different output results for the same task, which are then sorted or rated by human annotators based on preference criteria such as accuracy, fluency, professionalism, and alignment with values. Using this data with artificial preference labels, train a reward model (RM) that can automatically evaluate the quality of the model output, replacing manual large-scale preference judgments. For example, for different text interpretations of "The Cultural Influence of Slow Rhythm Intangible Cultural Heritage Music", the annotator sorts them according to "academic rigor" and "logical completeness", and the reward model can automatically score them after learning.

3. RL Fine-Tuning uses the score of the reward model as the "reward signal", and iteratively optimizes the supervised fine tuned model using reinforcement learning algorithms such as PPO and proximal strategy optimization. The model will continuously adjust its output strategy to maximize the reward score given by the model, ultimately generating results that are more in line with human needs.

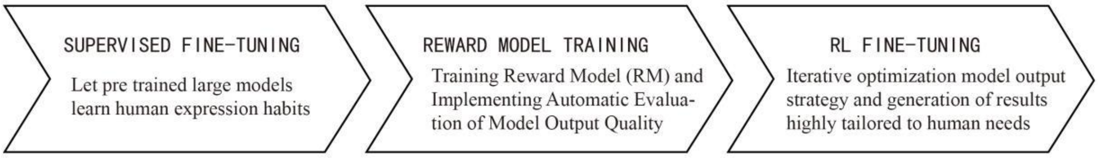

Figure 1 The Three Stages of Reinforcement Learning from Human Feedback.

4.1.3 Develop a cultural sensitivity detection module

For intangible cultural heritage elements such as Nuo noodles, shamanic drums, and ancestral totems that are sacred or taboo, a ban or strong review mechanism should be established. When the user prompt contains such keywords, the system automatically pops up a warning saying "This content involves sensitive symbols of ethnic culture, please confirm the purpose and obtain community authorization". At the same time, using image recognition technology to scan and generate results, if unauthorized secret patterns are found, they will be immediately blocked from publication. This type of 'technological ethical barrier' can effectively prevent cultural abuse.

4.2 Institutional Governance

Technical governance requires institutional safeguards. The current intangible cultural heritage AI urgently needs to establish a full cycle regulatory framework covering pre event, during event, and post event management

4.2.1 Clearly define prohibitive clauses

If it is not allowed to generate religious symbols that are detached from the ritual context, to fabricate lost secret techniques, and to use ethnic minority epics for commercial entertainment. At the same time, a "white list" mechanism is set up to encourage innovative experiments in open projects such as New Year pictures, Paper Cuttings, folk songs, etc. Jointly developed by cultural and tourism departments, representatives of inheritors, technical experts, and ethical scholars to ensure professionalism and representativeness.

4.2.2 Explore intellectual property protection mechanisms.

The digital wave has injected new vitality into the dissemination of intangible cultural heritage, but it has also brought deep challenges to the distribution of cultural discourse power and the ownership of rights and interests. As the crystallization of collective wisdom, the digital transformation process of intangible cultural heritage often involves the participation of multiple subjects, and the implicit intervention of power structures may lead to the neglect of the core interests of the inheriting group. For example, Taylor Taylor focused on analyzing how digitization enhances the discourse influence of cultural heritage at the level of power, exploring the role of this process in presenting public values, and exploring the functions of social media in it. He emphasized the need to deeply grasp the roles that digitization and social media play in the democratization process of cultural heritage[8].

4.2.3 Promote cultural impact assessment.

Before the launch of major intangible cultural heritage AI projects, a mandatory Cultural Impact Assessment (CIA) review should be conducted to assess their potential impact on cultural authenticity, community identity, and intergenerational inheritance. The evaluation report needs to be publicly solicited for opinions, especially feedback from inheritors and youth groups. Those who fail the assessment are not allowed to enter public communication channels. This move can shift risk prevention and control from post correction to prevention.

4.3 Subject Coordination

The ultimate destination of technology and institutions is people. Only by activating the initiative of diverse subjects can sustainable governance be achieved. Therefore, the primary task is to empower inheritors to master AI tools. Currently, most inheritors are elderly and have a fear of digital technology. We should develop AI assisted platforms with low threshold and dialect support, such as "automatic generation of pattern sketches" and "intelligent recognition process steps". At the same time, a training program for "digital inheritors" will be launched to teach skills in prompt word engineering, content review, and copyright management, transforming them from "passive digitization" to "active control of intelligence".

Secondly, establish the ultimate decision-making power of the inheritor. All AI generated intangible cultural heritage content published to the public, whether produced by platforms, users, or researchers, must be certified by relevant inheritors or their authorized institutions. A regional "Intangible Cultural Heritage AI Content Certification Center" can be established, with a review team composed of senior inheritors to conduct cultural compliance checks on the generated plans. Unverified content shall not be labeled with words such as "traditional" or "intangible cultural heritage" to prevent misleading the public.

Furthermore, promote the construction of community governance mechanisms. Intangible cultural heritage belongs to a specific community, and its digital destiny should not be left to technology companies to decide. We should encourage the establishment of committees composed of inheritors, village representatives, young makers, scholars, and developers to regularly review proposals for AI intangible cultural heritage projects, allocate data benefits, and develop community norms.

Finally, enhance the public's intelligence literacy and cultural recognition. The education department can add a module on "AI and Traditional Culture" to the aesthetic education curriculum of primary and secondary schools, guiding students to understand the limitations of AI generated content and learn to distinguish between real inheritance and technological simulation. Media platforms should also mandate the labeling of "AI generated" and provide a brief explanation: "This content is algorithmic creation and does not represent real intangible cultural heritage practices. Building a social defense against cultural distortion through the improvement of public literacy.

In summary, generative artificial intelligence cannot rely on a single approach for risk management of intangible cultural heritage. Only by building a solid bottom line through trustworthy technology, delineating ethical boundaries, and activating endogenous driving forces through subject collaboration, can we embrace technological innovation while preserving the spiritual roots of intangible cultural heritage. This is not only a technical issue, but also a historical proposition for the inheritance of civilization.

5. Conclusions

Generative artificial intelligence has brought new opportunities for the inheritance and innovation of intangible cultural heritage, but it has also quietly planted multiple hidden dangers. These issues directly point to the inheritance and authenticity of intangible cultural heritage. The key to cracking lies in the deep collaboration between technology, institutions, and stakeholders. But we need to be vigilant at all times, AI should be an auxiliary rather than a dominant force. The core of intangible cultural heritage inheritance has always been the cultural spirit carried by craftsmanship, sound, and memory, as well as the craftsmanship spirit of human beings as the main body of intangible cultural heritage inheritance. Only by adhering to the principle of putting people first and safeguarding cultural authenticity can tradition continue to thrive and rejuvenate in the wave of intelligence.

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